MOSES MAINA ONGWARE



Systems-Driven Software Engineer

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SUMMARY

Systems-driven software engineer, experienced in building reliable custom software, secure APIs, and production-ready full-stack systems. Strong in software architecture, backend engineering, frontend development, database design, system integrations, performance optimization, and automated testing. Uses systems thinking and AI-assisted workflows to clarify requirements, evaluate tradeoffs, and deliver maintainable software that solves complex operational problems.

PROFESSIONAL EXPERIENCE

Software Engineer | Eclectics International Limited — August 2026 – Present

- Architected bidirectional Maker-Checker role segregation and Four-Eyes Principle lockouts across Addis Banks' AML Portal, preventing self-authorization and ensuring immutable dual audit trails for financial crime compliance.
- Collaborated with DevSecOps and my Supervisor to resolve API contract drift and align request schemas, successfully unblocking and closing critical bank UAT defect cycles.
- Eliminated silent submission failures by refactoring Angular 18 modal lifecycle to retain form state and surface actionable HTTP 400 errors, backed by Jest unit tests using synthetic mock fixtures.

IT Support Trainee | *Rural Electrification & Renewable Energy Corporation (REREC)* — May 2024 – Aug 2024

- Supported SAP S/4HANA onboarding and troubleshooting for enterprise users.
- Managed Jira service requests covering hardware maintenance and IT operations.
- Monitored server-room infrastructure and assisted with device and network issue resolution.

PROFESSIONAL PROJECTS

WhatsApp Product Verification (WPV) | WPV Demo

- Developed a Node.js/TypeScript anti-counterfeiting platform supporting large-scale WhatsApp product verification, with an operations dashboard for code management, investigations, analytics, and reporting.
- Prevented duplicate activations and race-condition fraud using PostgreSQL transactions, row-level locking, automatic blocking, audit trails, JWT authentication, HMAC-SHA256 webhook validation, and rate limiting.
- Achieved sub-24ms p95 response times with zero failed requests during load testing at 200 concurrent users through indexing, query optimization, and connection pooling.

Rental Management System | Admin Demo | Landlord Demo | Tenant Demo | Caretaker Demo

- Architected a multi-tenant Property platform comprising four independently deployable React/TypeScript portals and a role-scoped Node.js/Express API for administrators, landlords, tenants, and caretakers.
- Engineered race-safe M-Pesa rent reconciliation using idempotent callbacks, Serializable transactions, PostgreSQL advisory locks, unique receipt constraints, and missed-callback recovery to prevent duplicate credits and double-billing.
- Built rent-cycle accounting, instalment plans, maintenance workflows, automated reminders, analytics, and asynchronous PDF/Excel reporting using PostgreSQL, Prisma, Redis, and BullMQ.
- Secured the platform with RS256 authentication, refresh-token rotation and revocation, RBAC, rate limiting, account-status enforcement, and audit logging.
- Maintained 263 passing backend tests across 67 suites, covering authentication, authorization, payment reconciliation, rate limiting, exports, and role-specific workflows.

EDUCATION

Bachelor of Science in Computer Science | **Chuka University** — Graduated Nov 2025

Relevant Coursework: Software Engineering, User Interface Design, Distributed Systems, Decision Support Systems, Component Based Architecture, Software Project Management, Artificial Intelligence, Data Structures.

TECHNICAL STACK

Languages: TypeScript, Python, SQL, PHP. Java.
Frontend: React, Angular, Next.js, Vue.js, TanStack Query, React Router, Vite.
Backend & APIs: Node.js, Express.js, Laravel, Yii3, REST API Development, Webhooks, Background Jobs, JWT Authentication, RBAC, BullMQ.
Databases & Data: PostgreSQL, MongoDB, Redis, Prisma ORM, Database Design, Transactions, Query Optimization, Caching.
Integrations: M-Pesa Daraja API, Meta WhatsApp Cloud API, Cloudinary, Resend, OAuth, OpenSMS.
Tests & Quality: Jest, Supertest, Playwright, k6, Integration Testing, Load Testing, API Testing, Codeception(PHP).
DevOps & Cloud: Docker, GitHub Actions, Linux, Git, GitHub, Vercel, Render, Prometheus, Winston, Pino, ngrok, VPS deployment.
Engineering & Competencies: Software Architecture, Systems Thinking, Custom Software Development, Backend Engineering, API Design, Security Engineering, Performance Optimization, Technical Problem Solving.

CERTIFICATION

Docker Bootcamp for ML, GenAI and Agentic AI • Prompt Engineering with Amazon Q Developer • Cybersecurity & Emerging Technologies Awareness